
O7 SOLUTIONS
Quality in Everything

Machine Learning & Deep Learning Internship Test

Instructions

You are required to complete all mandatory sections of the test. Make sure to submit your code for each question. Your answers will be evaluated based on correctness, logic, model selection, evaluation, and efficiency. You may use Google Colab with GPU acceleration. You are free to use any Python-based machine learning or deep learning framework. You may use official documentation and online references. Do not use ChatGPT or any other AI tools to complete the tasks. Your submission must reflect your own work. Additional insights and meaningful improvements are encouraged.

Duration: 90 Min.

MACHINE LEARNING

Dataset: Cardiovascular Disease Risk Prediction Dataset

Dataset link:

https://www.kaggle.com/datasets/harshwardhanfartale/cardiovascular-disease-risk-prediction-dataset/data

Objective

Build a complete classification workflow to predict whether a patient is at risk of cardiovascular disease. Your solution should demonstrate practical understanding of preprocessing, classification, ensemble learning, model evaluation, and optimization.

1. Data Understanding & Preparation

Load and inspect the dataset. Identify the target variable and feature types. Check missing/null values and duplicate records. Inspect the target-class distribution. Identify and handle relevant data-quality issues. Prepare the data for machine learning.

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2. Preprocessing Pipeline

- Perform an appropriate train/test split.
- Create a reusable preprocessing pipeline for numerical and categorical features.
- Apply suitable encoding and scaling where required.
- Ensure that your preprocessing approach does not cause data leakage.

3. Baseline Classification Model

- Train a baseline classification model. Evaluate it using Accuracy, Precision, Recall, F1-score, and a Confusion Matrix. Briefly record your observations.

4. Support Vector Machine (SVM)

- Train an SVM classifier using the preprocessing pipeline. Experiment with suitable hyperparameters such as C, kernel, and gamma where applicable. Evaluate the model using appropriate classification metrics. Generate the ROC curve and calculate ROC-AUC.

5. Ensemble Learning

- Implement a Bagging-based classification model.
- (Optional) Implement a Boosting-based classification model.
- (Optional) Train an AdaBoost classifier separately.
- Evaluate and compare the ensemble models using suitable metrics.

6. Model Optimization

- Select at least one model for optimization. Use one or more suitable techniques such as Grid Search, Randomized Search, Cross-Validation, class weighting, threshold tuning, or feature selection. Show the performance before and after optimization. Do not optimize only for accuracy; justify the metrics used.

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7. Advanced Model Evaluation

- Compare Precision, Recall, F1-score, and ROC-AUC for your important models.
- Plot and compare ROC curves where appropriate.
- Investigate how changing the classification threshold affects model predictions.
- Comment on false positives and false negatives in the context of cardiovascular disease risk.

Optional / Additional Insights

- Precision-Recall curve
- Class imbalance handling
- Feature importance or model interpretability
- Probability calibration
- SHAP or another explainability method
- Additional hyperparameter experiments

DEEP LEARNING

Dataset: AQUA20 Underwater Species Classification

Dataset link:

https://huggingface.co/datasets/taufiktrf/AQUA20

Classes: 20 underwater species classes

Objective

Build an image classification model that can classify unseen real-world underwater images. You may use transfer learning and a pretrained CNN architecture. Building a CNN from scratch is not required.

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8. Image Dataset Preparation

- Load and inspect the image dataset. Create suitable training, validation, and testing sets. Resize and normalize the images appropriately. Inspect the class distribution. Apply suitable image preprocessing.

9. Transfer Learning Model

- Use a suitable pretrained CNN architecture for image classification. Train the model on the provided dataset. Record training and validation accuracy and loss. Plot the training and validation curves.

10. Model Optimization

- Analyze the training and validation curves for signs of overfitting or underfitting. Apply at least one meaningful optimization technique. Possible techniques include data augmentation, dropout, learning-rate adjustment, early stopping, batch-size adjustment, or fine-tuning pretrained layers. Compare the original and optimized models.

11. Performance Testing

- Evaluate the final model on unseen test data.
- Report test accuracy and suitable per-class performance metrics.
- Generate a confusion matrix.
- Display several test images with their actual and predicted classes.
- Identify at least one type of image or class that appears difficult for the model to classify.

Submission Requirements

- Submit the completed Google Colab/Jupyter notebook.
- Ensure all important outputs, plots, metrics, and observations are visible.
- Keep the notebook organized and readable.
- Ensure the code can be understood and executed by another person.

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